



Thermomechanical-induced cracking model for ceramic matrix composite laminates subjected to thermal gradients and transients

Zhengmao Yang^{a,*}, Jiangtao Wang^b, Rui Yang^c, Jian Jiao^{c,*}

^a Institute of Mechanics, Chinese Academy of Sciences, Beijing, 100190, China

^b School of Aerospace Engineering, Beijing Institute of Technology, Beijing 100081, China

^c National Key Laboratory of Advanced Composites, AECC Beijing Institute of Aeronautical Materials, Beijing 100095, China

ARTICLE INFO

Keywords:

Ceramic-matrix composites (CMCs)

Failure

Energy release rate

Transient thermal gradients

ABSTRACT

Regarding the potential damage and failure issues that may occur in high-temperature transient environments during the service process of ceramic matrix composite laminates, the present work proposes a multi-layer thermomechanical induced crack initiation model for ceramic matrix composite laminates subjected to thermal gradients and transients. This model can determine the history of temperature, deformation, and stress distribution within each layer of the material, as well as the steady-state energy release rate of all possible crack locations. Then the influence of bending constraints on material stress distribution and energy release rate is investigated, finding that maximum stress and energy release rate values are significantly lower without bending constraints compared to those under bending constraints. Furthermore, the study explores the impact of material structural parameters and boundary conditions on the energy release rate of ceramic matrix composite laminates, providing direct insights for designing laboratory tests and evaluating the lifespan of these materials under in-service conditions.

1. Introduction

Ceramic Matrix Composites (CMC) are increasingly preferred over traditional high-temperature alloys due to their high damage tolerance and specific strength at elevated temperatures. This has led to their adoption in critical applications such as thermal protection systems for hypersonic reentry vehicles (Hermann et al., 2019; Di Caprio et al., 2021), hot end components of aircraft engines (Padture, 2016; Zok et al., 2021), and nuclear reactor liners (Alabdullah and Ghoniem, 2020; Bumgardner et al., 2021). However, the severe operating conditions in these applications demand that CMC withstand very high temperatures, high thermal flux with significant transient characteristics, thermal/mechanical shock, high-temperature oxidation, and fatigue. The nonlinear transient thermal mechanical response arises from the strong coupling effect of transient thermal stress due to local high temperature gradients and mechanical stress from mechanical loads, high temperatures, and complex microstructure deformation of the material. As a result, damage exhibits multi-scale evolution and multi-mode coupling, presenting significant challenges for the reliable design and safe operation of CMC structures in high-temperature transient environments (Lu and Fleck, 1998; Hutchinson and Evans, 2002; Jackson et al., 2015).

During the re-entry stage of hypersonic vehicles, severe transient conditions cause high cyclic thermal stress that exceeds internal constraints compared to steady-state flight. This is due to the rapid increase in leading-edge temperature caused by thermal convection from high heat flux. CMC systems have low thermal conductivity, resulting in a temperature gradient between the thinner leading edge and the thicker main body. As the temperature of the leading edge expands, it encounters resistance from the main body with a lower temperature, resulting in tensile load on the main body while the leading edge experiences compression. Once steady-state conditions are reached, the transient stresses disappear. However, the thermal cycle promotes low cycle fatigue, which can be detrimental to the lifespan and reliability of the hypersonic vehicle (Glass, 2008; Pichon et al., 2009).

When materials are subjected to transient thermomechanical loads, their response involves non-equilibrium phenomena such as strengthening, softening, multi-scale damage, and energy dissipation. These complex mechanisms pose significant challenges for researchers attempting to establish corresponding mechanical theories and methods. The highly nonlinear nature of transient thermomechanical loading, coupled with the intricate microstructure of materials, makes it difficult to accurately predict the response of materials under these conditions. Moreover, the presence of multiple scales of damage and energy

* Corresponding authors.

E-mail addresses: zmyang@imech.ac.cn (Z. Yang), Jian.Jiao@biam.ac.cn (J. Jiao).

<https://doi.org/10.1016/j.ijsolstr.2024.112927>

Received 15 February 2024; Received in revised form 25 April 2024; Accepted 11 June 2024

Available online 13 June 2024

0020-7683/© 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

dissipation mechanisms further complicates the problem. Researchers have developed various approaches to study the response of materials under transient thermomechanical loads, including experimental testing, numerical simulations, and theoretical modeling. Sundaram et al. (2012) presented a systematic study of factors affecting the delamination energy release rate and mode mix of a thermal barrier coating attached to a substrate accounting for the influence of thermal gradients combined with rapid hot surface cooling. Subsequently, Jackson and Begley (2014) extended the research on the impact of transient effects mentioned above, presented critical cooling rates to avoid cracking in thermal barrier coatings (TBCs) driven by thermal transients, and analyzed bilayers over a broad range of properties and cooling rates. Koutsakis et al. (2022) investigated the durability of thermal barrier coatings for reciprocating internal combustion engine applications using a coupled, unsteady fracture-based thermomechanics model. The existing work provides a foundational understanding of the thermomechanical behavior of materials subjected to high-temperature transient conditions. Nevertheless, given the multi-component nature and the inherent heterogeneity and anisotropy of CMC laminates, it is imperative to expand the current analytical framework to comprehensively investigate and elucidate the damage mechanisms within CMC laminates, particularly under the influence of thermal gradients and transient thermal conditions. Yang and Yuan (2019) revealed the thermo-mechanical damage mechanism of oxide/oxide CMC through a series of cyclic thermal shock tests, and established corresponding thermal mechanical damage evolution models based on continuum damage mechanics (CDM) and irreversible thermodynamic theory. Hattiangadi and Siegmund (2004) described the coupled thermo-mechanical numerical analysis of composite laminates with bridged delamination cracks loaded by a temperature gradient, and presented the traction-separation law which accounts for breakdown of the micromechanisms responsible for load transfer across bridged delamination cracks. Shi et al. (2023) proposed a thermo-mechanical strength-based cohesive zone model to simulate the behavior of an imperfect interface, in which the load transfer behavior is described by the strength model and the heat transfer behavior is delineated by the interface conductance including the bonding conductance. Despite this, significant challenges remain in accurately capturing the complex phenomena occurring at different scales and predicting the behavior of materials under these conditions.

The primary objective of this work is to address the issue of transient temperature-induced stresses in CMC laminates and how these stresses contribute to thermo-mechanical damage, such as cracks or interface delamination. To address this objective, a brief description of transient thermal-induced damage was introduced in Section 2 firstly. Then, a computational framework was established to characterize the transient temperature and stress distribution inside CMC laminates during rapid heating processes in Section 3. Furthermore, the proposed framework was verified by experiments as described in Section 4. Finally, the transient energy release rate of CMC stratification was determined, and conducted parameter analysis on several factors that affect the energy release rate of materials in Section 5. This study provides guidelines for better understanding the failure mechanisms driven by transient conditions, as well as acceptable transient thermal conditions in practical applications.

2. Brief description of the transient thermal-induced damage

In high-speed flight thermal protection systems, the CMC laminates is exposed to severe aerodynamic and aerothermal effects, especially the transient high heat flux with thermal gradients. The flow of heat through the multilayer in transient heating scenarios occurs much faster than the change in surface temperatures. Due to the low thermal conductivity of CMC, the unbalanced aerodynamic heat input on the surface of the structure quickly accumulates, causing a sharp increase in temperature and presenting a 'micro-time scales induced transient'

characteristic. Additionally, the composite structure demonstrates a 'high temperature gradient' across its thickness.

As shown in Fig. 1(a), the temperature distribution along the thickness of the structure can be divided into three regions: (i)Region I, located close to the outer surface of the structure, experiences the largest temperature gradient and the highest temperature; (ii)Region II represents the middle part of the structure, where the thermal gradient gradually flattens; (iii)Region III is near the inner surface of the structure, characterized by relatively lower temperature and less noticeable thermal gradient. These observations highlight the distinct temperature profiles and thermal characteristics within the CMC laminates during transient heating situations. Understanding these temperature distributions and gradients is crucial for accurately assessing the response and potential damage mechanisms of the material under such conditions.

The dual coupling effect of 'micro-time scales induced transient' and 'high temperature gradient' can cause instantaneous and uneven stress distribution in the CMC. Previous studies (Wang and Yang, 2024a,b) have indicated that stresses and strain energy generated during transients can be more damaging than those associated with quasi-steady-state temperature distributions resulting from slow heating processes. In Region I, where a high thermal gradient exists, the induced transient stress can have two significant consequences:(i) matrix enriched micro-region cracks and growth under tensile stress, and (ii) fiber/matrix interface micro-region debonding within the layer, as well as delamination and edge delamination of surface cracks under shear stress caused by coefficient of thermal expansion (CTE) mismatch or additional stress. These observed phenomena are demonstrated in Fig. 1(b). The presence of these localized cracks, debonding, and delamination greatly compromises the structural integrity of the CMC laminates, eventually leading to overall structural failure under aerodynamic forces. Understanding the mechanisms behind these stress-induced failure modes is essential for developing strategies to mitigate or prevent damage in CMC laminates exposed to transient thermal conditions.

Therefore, The consideration of thermal transients and large thermal gradient effects is crucial when assessing the reliability of CMC laminates. Traditional design criteria for structural thermal strength typically focus on quasi-steady-state high-temperature loads, which only address material performance degradation once the structural temperature stabilizes (Parthasarathy et al., 2004). These criteria often overlook the significant impact of micro-time scales induced transient and high-temperature gradients on material behavior. When discussing the effects of thermal transients on materials, two scenarios are commonly considered: (i) Thermal Shock in an Unconstrained Uniform Semi-Infinite Solid: In this scenario, when the temperature of the entire free surface of a uniform semi-infinite solid suddenly increases, a thermal shock is generated, leading to compressive stress parallel to the surface. If there exists a weak interface parallel to the surface, sub-surface mixing mode stratification may occur, impacting the material's integrity. (ii) Localized Temperature Increase in a Uniform Semi-Infinite Solid: In this case, when a higher temperature is suddenly applied to a local region of a uniform semi-infinite solid, it can result in localized stress concentrations and gradients within the material, potentially leading to deformation, cracking, or other forms of damage.

Taking into account these scenarios and understanding the effects of thermal transients is essential for accurately predicting the behavior and performance of CMC laminates under transient thermal conditions. Failure to consider these factors may result in misleading assessments of the reliability and structural integrity of the materials.

3. General model framework

Building upon the analytical approaches of prior research (Sundaram et al., 2012; Jackson and Begley, 2014), which scrutinized the layering mechanisms of multilayer materials through an energetic perspective, the present work extends the investigation to address the

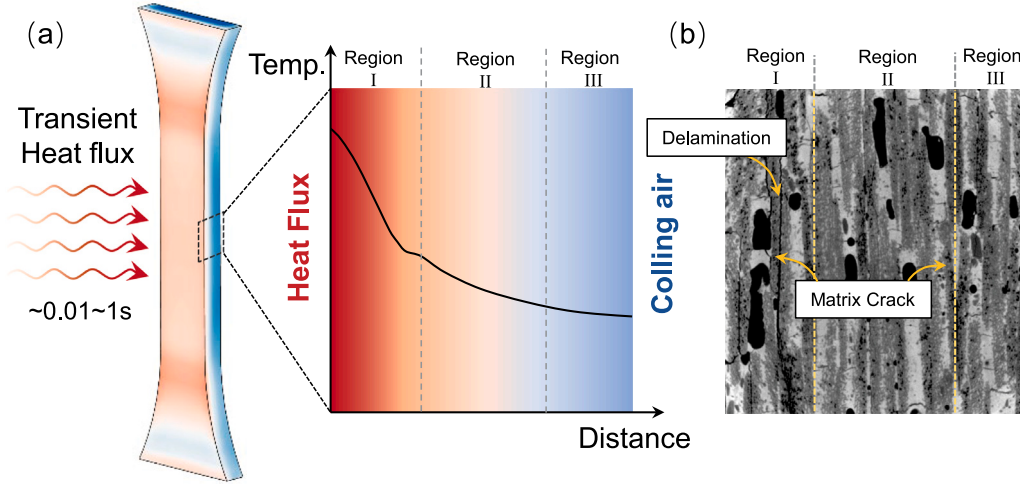


Fig. 1. (a) The temperature distribution along the thickness of the CMC laminate subjected to thermal gradients and transients, and (b) The thermomechanical damage distribution along the thickness.

novel challenges posed by CMC laminates under transient thermal gradients. To this end, a novel computational framework has been formulated. This framework is specifically tailored to elucidate the damage and failure mechanisms of CMC laminates when subjected to transient high-temperature gradients, which aims to establish the relationship among the temperature rise rate, the delamination damage location as well as the moment the damage occurs, and can predict the influence of non-uniform temperature fields generated by various actual operation conditions on material properties. The model of CMC laminates under transient thermal loads consists of two parts: transient thermal analysis and subsequent thermal coupling analysis.

3.1. Basic assumption

The model framework adopts the following basic assumptions:

- (i) The thickness length scale of CMC laminates is much smaller than the length scale of the material in both directions within the plane. Therefore, the model assumes that material temperature and deformation variations occur only in the through-thickness direction. This means that temperature, strain, and stress can be expressed as functions of material thickness, denoted as $T = T(z)$, $\epsilon = \epsilon(z)$ and $\sigma = \sigma(z)$, respectively, and the material always maintains a plane stress state during deformation.
- (ii) In the initial state, the temperature distribution inside the material is uniform, and there are no residual stresses. The temperature distribution of materials depends solely on the given one-dimensional heat conduction problem and is not affected by their deformation state.
- (iii) The time scale required for material deformation and layered crack propagation to reach a steady state is much smaller than the time scale of temperature changes associated with delamination cracking. Therefore, the material's deformation state under any transient temperature distribution depends only on the current temperature distribution and is independent of the temperature time gradient.

3.2. Inelastic constitutive model of CMC laminates

In the current work, a nonlinear constitutive model is employed to characterize the mechanical behavior of CMC, which is more closely approximates real-world service conditions. Upon establishing this nonlinear model, it becomes feasible to ascertain the space-time evolution

of the entire stress field. The obtained stress field is fundamental to the computation of the steady-state energy release rate. The inelastic constitutive model describes the in-plane strains as functions of in-plane stresses in the primary axes (the two fiber directions) using a phenomenological model based on the deformation theory of plasticity (Genin and Hutchinson, 1997). The strains for composite laminates in the original formulation of the model are calculated via an effective compliance matrix of the form,

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma \end{bmatrix} = \begin{bmatrix} S_{11}(\bar{\sigma}) & S_{12}(\bar{\sigma}) & 0 \\ S_{21}(\bar{\sigma}) & S_{22}(\bar{\sigma}) & 0 \\ 0 & 0 & S_s(\bar{\sigma}) \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau \end{bmatrix} \quad (1)$$

where σ_1 , σ_2 , and τ are the normal and transverse stress in strain in the fiber directions and shear stress, respectively. ϵ_1 and ϵ_2 are axial and transverse strains as functions of axial stress in a tensile test in the $0^\circ/90^\circ$ orientation, γ is the shear strain in a $0^\circ/90^\circ$ shear test. $\bar{\sigma}$ is a Hill-type effective stress relates the stresses in the $0^\circ/90^\circ$, $\pm 45^\circ$, and shear configurations, defined by Rajan et al. (2014)

$$\bar{\sigma} = \sqrt{A_{11}(\hat{\sigma}_1^2 + \hat{\sigma}_2^2) + A_{12}\hat{\sigma}_1\hat{\sigma}_2 + A_s\sigma_s^2} \quad (2)$$

where the ramp function $\hat{\sigma} = (|\sigma| + \sigma)/2$ is used to prevent compressive inelasticity. Through Eq. (2), the von Mises effective stress can be extended to anisotropic materials.

To obtain the constitutive model of a specific CMC laminate, three sets of standard mechanical tests need to be conducted on the composites, uniaxial tensile test in the fiber directions, uniaxial tensile test in the 45° with the fiber direction, and pure shear test in the fiber direction. Through these experiments, the failure stress of the material in three corresponding directions can be obtained σ_0 , σ_{45} and σ_s .

Under three sets of uniaxial loading conditions, using the definition of the effective stress in Eq. (2), the effective stress values of material failure are

$$\begin{aligned} C_0 &= \sqrt{A_{11}} \\ C_{45} &= \frac{1}{2} \sqrt{2A_{11} + 2A_{12} + A_s} \\ C_s &= \sqrt{A_s} \end{aligned} \quad (3)$$

where C_0 , C_s , and C_{45} are constants that need to be the constitutive model, which are determined through experiment calibration. Subscripted numerical values (0, 45) represent the smallest angle between the loading direction and the fiber direction, while the stresses, σ_0 , σ_{45} , and σ_s , are the cracking stresses, or proportional limits, for the three respective loadings. The latter constants nominally relate the stresses at

the onset of nonlinearity for $0^\circ/90^\circ$ tension (σ_0), pure shear (σ_0) and 45° tension (σ_{45}), via

$$C_0\sigma_0 = C_{45}\sigma_{45} = C_s\sigma_s \quad (4)$$

One of the constants, C_0 , is arbitrarily set to 1. Using Eq. (4), the constants A in the effective stress function can be evaluated (up to an arbitrary scale factor). The model assumes plane stress, cubic in-plane symmetry of composite properties, monotonic and proportional loading, and no shear/normal coupling. To ensure consistency between the constitutive equation and experimental results,

$$S_{11}(\bar{\sigma}) = \frac{f_0(\bar{\sigma}/C_0)}{\bar{\sigma}/C_0}, S_{12}(\bar{\sigma}) = \frac{f_{0T}(\bar{\sigma}/C_0)}{\bar{\sigma}/C_0}, S_{33}(\bar{\sigma}) = \frac{f_s(\bar{\sigma}/C_s)}{\bar{\sigma}/C_s} \quad (5)$$

where $\varepsilon_0(\sigma)$ and $\varepsilon_{0T}(\sigma)$ characterizes the experimental axial and transverse strain induced by tensile stressing in the composite in the $0^\circ/90^\circ$ orientation, and $\varepsilon_s(\sigma)$ is the experimental in-plane shear strain induced by shear stressing. Therefore, based on experimental data, f_0 , f_{0T} and f_s , the constitutive equation of the material can be obtained as follows:

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma \end{bmatrix} = \begin{bmatrix} \frac{f_0(\bar{\sigma}/C_0)}{\bar{\sigma}/C_0} & \frac{f_{0T}(\bar{\sigma}/C_0)}{\bar{\sigma}/C_0} & 0 \\ \frac{f_{0T}(\bar{\sigma})}{\bar{\sigma}/C_0} & \frac{f_0(\bar{\sigma})}{\bar{\sigma}/C_0} & 0 \\ 0 & 0 & \frac{f_s(\bar{\sigma}/C_s)}{\bar{\sigma}/C_s} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau \end{bmatrix} \quad (6)$$

3.3. Transient thermal analysis

In transient heat transfer analysis, it is necessary to determine the transient temperature distribution $T(z, t)$ at any time in the material through the initial temperature distribution, thermal boundary conditions, and various thermal physical parameters of the material.

CMC laminates are multi-layer structures, and the thickness, thermal conductivity, and thermal diffusion rate of each layer of material are denoted as h_i , κ_i and k_i . In the initial state, it is assumed that the temperature distribution of the material is known and steady, i.e. $T(z, 0) = T_0(z)$. The various thermal physical parameters of materials are affected by temperature, especially in high-temperature environments, which cannot be ignored. Therefore, the thermal diffusion rate and thermal conductivity coefficient of materials are also expressed as functions of material temperature, i.e. $\kappa_i = \kappa_i(T(h, t))$ and $k_i = k_i(T(h, t))$.

For different experiments or practical applications, the actual heat transfer forms of materials vary, but the thermal boundary of materials can be uniformly simplified as a forced convection heat transfer model. The forced convection heat transfer coefficient on the surface of the material can be obtained through experimental measurements. In this article, only the case related to the surface temperature of the material is currently considered, i.e. $H = H(T(h, t))$; For other cases where the heat transfer coefficient depends on more parameters, the method proposed in this article can be extended. The heat transfer coefficients between the upper and lower surfaces of the material and the outside world are denoted as H_h and H_0 respectively, and the temperature of the gas that undergoes convective heat transfer with the material is denoted as $T_h^{gas}(t)$ and $T_0^{gas}(t)$.

In summary, the transient temperature distribution of materials can be attributed to the following definite solution problem, the boundary conditions for material structures can be written as:

$$\begin{aligned} k_1(T) \frac{\partial T(0, t)}{\partial z} &= -H_0 (T_0^{gas}(t) - T(0, t)) \\ k_n(T) \frac{\partial T(h, t)}{\partial z} &= H_h (T_h^{gas}(t) - T(h, t)) \end{aligned} \quad (7)$$

Internal control equation for single-layer materials is,

$$\rho c(T) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left(k(T) \frac{\partial T}{\partial z} \right), \quad \frac{(i-1)h}{n} \leq z \leq \frac{ih}{n} \quad (8)$$

3.4. Stress analysis induced by rapid heating

Based on the temperature distribution obtained in Section 3.3, the stress distribution will be solved for the three constraint situations: unconstrained, constrained by bending on one side, and constrained by bending on both sides. Each scenario has a unique strain function. The coefficient of thermal expansion (CTE) of each material layer, denoted as $\alpha_i(T)$, is a function of temperature. It should be noted that the analysis focuses solely on changes in stress distribution caused by aerodynamic heating and does not consider residual stress in the initial state. The true deformation of materials can be discussed in three situations, as shown in Fig. 2: (i) Bilateral bending constraint, (ii) Unilateral bending constraint, and (iii) No bending constraint. In these three cases, the strain function of the material can be expressed in different forms. Depending on the experimental or actual usage environment, a suitable scenario can be used to analyze and calculate the deformation of the material.

3.4.1. Bilateral bending constraint

In the region located ahead of the delamination crack within the material, the mechanical behavior can be considered equivalent to that of intact materials. When subjected to double-sided bending constraint, the strain distribution in the thickness direction of the material remains uniform at all times. Therefore, the strain function of the material can be expressed as follows:

$$\varepsilon = \varepsilon(z, t) = \varepsilon^0(t) \quad (9)$$

Correspondingly, the mechanical strain of the material is:

$$\varepsilon^M(z, t) = \varepsilon^0(t) - \alpha(T)(T(z, t) - T(z, 0)) \quad (10)$$

Therefore, the stress distribution of the material is:

$$\sigma(z, t) = \Sigma(\varepsilon^M(z, t)) \quad (11)$$

Under the constraint of double-sided bending, the material should satisfy the following force balance equation:

$$\begin{aligned} \int_0^h \sigma(z, t) dz &= \int_0^h \Sigma(\varepsilon^M(z, t)) dz \\ &= \int_0^h \Sigma(\varepsilon^0(t) - \varepsilon^{th}(z, t)) dz = 0 \end{aligned} \quad (12)$$

Given the temperature distribution, thermal expansion coefficient, and constitutive equation of the material, the true strain and stress distribution of the material can be obtained by solving Eqs. (9)–(12).

Far behind the delamination cracks in the material, the delaminated portion can be treated as an independent entity. The time scale of delamination crack propagation is much smaller than the time scale of heat conduction. As a result, delamination crack propagation can be regarded as an instantaneous process, and the temperature distribution within the material remains unchanged before and after delamination.

In the case of layered materials, equilibrium equations can be established for each layer interface. Taking the example of $n-1$ interfaces in a multi-layered material, the equilibrium equations are as follows:

$$\varepsilon^M(z, t) = \varepsilon_i^0(t) - \alpha_i(T)(T(z, t) - T(z, 0)) \quad (13)$$

and

$$\int_{\frac{(i-1)h}{n}}^{\frac{ih}{n}} \Sigma(\varepsilon^M(z, t)) dz = 0. \quad (14)$$

3.4.2. Unilateral bending constraint

Far ahead of the delamination crack, the deformation state of the material is identical to that of a double-sided constraint. The stress and strain distribution can be determined by solving the following equation:

$$\varepsilon^M(z, t) = \varepsilon^0(t) - \alpha(T)(T(z, t) - T(z, 0)) \quad (15)$$

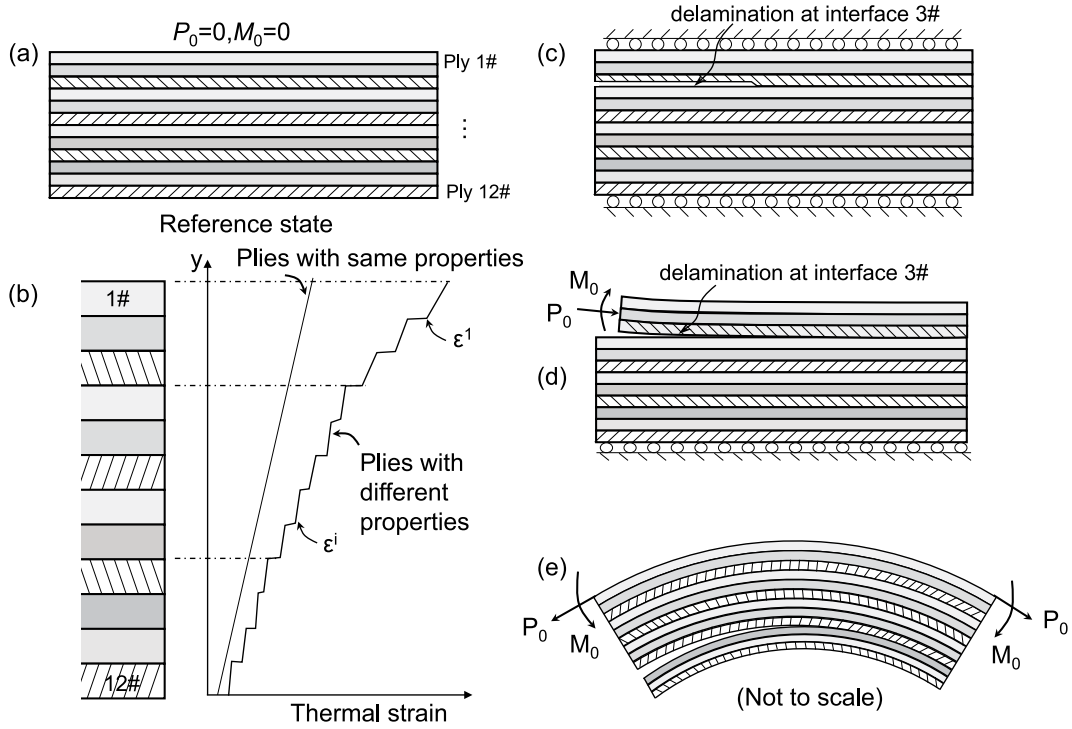


Fig. 2. (a) A CMC laminate is stress-free when no subjected to external thermomechanical loads, resulting in thermal strains $\epsilon_{th} = 0$; (b) Strains within each layer of the CMC laminate are defined as linear functions; (c) Bilateral bending constraint; (d) Unilateral bending constraint, and (e) No bending constraint.

and

$$\int_0^h \Sigma (\epsilon^M(z, t)) dz = 0 \quad (16)$$

Far behind the delamination crack, only a portion of the material that contains the constraint surface remains under bending constraints. The equilibrium equation for this region of the material is still:

$$\int_{thk} \Sigma (\epsilon^M(z, t)) dz = 0 \quad (17)$$

For the remaining materials that are no longer constrained by bending after delamination, their strain function follows the following form:

$$\epsilon = \epsilon(z, t) = \epsilon^0(t) + \rho(t)z \quad (18)$$

where $\rho(t)$ notes the curvature of the composite laminate.

For the layered part that is not constrained by bending, the balance equation should also include the torque balance equation,

$$\epsilon^M(z, t) = \epsilon^0(t) + \rho(t)z - \alpha(T)(T(z, t) - T(z, 0)) \quad (19)$$

and

$$\begin{aligned} \int_{thk} \Sigma (\epsilon^M(z, t)) dz &= 0 \\ \int_{thk} \Sigma (\epsilon^M(z, t)) z dz &= 0 \end{aligned} \quad (20)$$

By solving the above Eqs. (15)–(20), the strain distribution of the material can be obtained. It should be noted that for the layered part, if the curvature of the upper layer is smaller than that of the lower layer, the result obtained by solving the above equation is reasonable. If the curvature of the upper layer is greater than that of the lower layer, it indicates that even after layering, these two layers are still in contact with the same curvature and need to be resolved. The equilibrium equation for this scenario is as follows:

$$\begin{aligned} \epsilon_1^M(z, t) &= \epsilon_1^0(t) + \rho(t)z - \alpha(T)(T(z, t) - T(z, 0)) \\ \epsilon_2^M(z, t) &= \epsilon_2^0(t) + \rho(t)z - \alpha(T)(T(z, t) - T(z, 0)) \end{aligned} \quad (21)$$

and

$$\begin{aligned} \int_{thk1} \Sigma (\epsilon_1^M(z, t)) dz &= 0 \\ \int_{thk2} \Sigma (\epsilon_2^M(z, t)) dz &= 0 \end{aligned} \quad (22)$$

3.4.3. No bending constraint

For the scenario without bending constraints, the material must satisfy the balance equations of the principal vector and torque in front of the delamination crack. The equilibrium equations are as follows:

$$\epsilon^M(z, t) = \epsilon^0(t) + \rho(t)z - \alpha(T)(T(z, t) - T(z, 0)) \quad (23)$$

and

$$\begin{aligned} \int_0^h \Sigma (\epsilon^M(z, t)) dz &= 0 \\ \int_0^h \Sigma (\epsilon^M(z, t)) z dz &= 0 \end{aligned} \quad (24)$$

Behind the delamination crack, each part of the material after delamination needs to satisfy the balance equations of the principal vector and moment. For the parts that are still in contact after delamination, the equation needs to be re solved according to the method of solving material deformation under single-sided bending constraint.

3.5. Energy release rate for steady-state interface delamination

When a crack extends from the edge of the material or when there is an open vertical crack that is sufficiently long, the delamination at the material interface can reach a steady state. In this steady-state delamination state, the energy release rate of the material's delamination crack can be calculated by considering the energy change from the crack tip front (non-delaminated material) to the back (steady-state delaminated material) simultaneously.

Based on the assumption mentioned earlier, where the time scale for crack propagation is much smaller than the time scale for temperature changes, it can be assumed that during the rapid heating process, the

layered cracks in the material have fully developed into a steady state. Therefore, the difference in strain energy between the material before and after the crack tip can be considered as the strain energy release rate of the material.

The ratio of strain energy to area of the non-destructive material far ahead of the crack tip is:

$$U_{HD} = \int_0^{h_1+h_2} \left(\int \sigma d\epsilon \right) dz \quad (25)$$

The ratio of strain energy to area of the non-destructive material far ahead of the crack tip is:

$$U_{BH} = \int_0^{h_1} \left(\int \sigma d\epsilon \right) dz + \int_{h_1}^{h_1+h_2} \left(\int \sigma d\epsilon \right) dz \quad (26)$$

where σ is obtained by the nonlinear constitutive model. The energy release rate can be calculated from the definition for the strain energy following the same method presented in Sundaram et al. (2012) and Jackson and Begley (2014),

$$G = U_{HD} - U_{BH} \quad (27)$$

The steady-state energy release rate associated with a semi-infinite interface crack placed at an arbitrary location from Eq. (27) can be calculated numerically.

The mode mix, ψ is obtained using solutions for a crack lying along an interface in a bilayer. Consider the unconstrained case first. Define the force/thickness P and moments/thickness M acting on the layers due the stress in the uncracked bilayer by,

$$P = \int_{h_2} \sigma dz, M = \int_{h_1} \sigma \left(h_2 + \frac{1}{2} h_1 - (h - x) \right) dz \quad (28)$$

The parameter β_D is less important than α_D in the present problem, and it will be taken to be zero to simplify as:

$$\psi \equiv \tan^{-1} \left(\frac{K_{II}}{K_I} \right) = \tan^{-1} \left[\frac{\lambda \sin \omega - \cos(\omega + \gamma)}{\lambda \cos \omega + \sin(\omega + \gamma)} \right] \quad (29)$$

With these loads applied, a delamination crack can be introduced along the interface with no change in stress in the system. The stress intensity factors associated with delamination arise due to the removal of these loads to achieve the delaminated CMC composites. The following dimensionless parameters enter the evaluation of Eq. (28),

$$\begin{aligned} \eta &= h_1/h_2, \quad \Sigma = \bar{E}_2/\bar{E}_1, \\ A &= [1 + \Sigma(4\eta + 6\eta^2 + 3\eta^3)]^{-1}, \\ I &= [12(1 + \Sigma\eta^3)]^{-1}, \\ \sin \gamma &= 6\Sigma\eta^2(1 + \eta)\sqrt{AI}, \\ \lambda &= \sqrt{\frac{I}{A}} \frac{Ph_2}{M} \end{aligned} \quad (30)$$

No corresponding elasticity solution for the mode mix is available for the constrained bending problem. An infinitely deep substrate constrains bending, but it also constrains longitudinal straining of the substrate occurring during delamination. Of the two effects, the bending constraint is the more significant in determining the mode mix. As an approximation to the constrained bending case, the result (Eq. (28)) is used with $\eta \rightarrow 0$, corresponding to a coating delaminating from a deep substrate.

4. Composite material and assumptions in the present multilayers laminates analyses

4.1. Composite properties

To gain broad insights into the effects of transient high temperature, a CMC laminate, fabricated by the Beijing Institute of Aeronautical Materials (BIAM), has been selected for detailed modeling. The laminate consists of uncoated Nextel™ 720 fibers (3M, St. Paul, MN) in

Table 1

Properties used in the CMC laminate.

Thermophysical properties	RT	1473K
Thermal expansion [ppm/K]	4.7	7.1
Thermal conductivity [W/(m K)]	3.49	2.04
Specific heat capacity [J/(g K)]	0.82	1.27

an 8HSW, 0°/90° woven layer configuration. The composite has a density of approximately 2.64 g/cm³ and a fiber volume fraction of around 46.4%. The properties of the composites are listed in Table 1, which were obtained from the ASTM standard tests conducted by the manufacturer (BIAM). This Al₂O₃ fabric preform was infiltrated with an alumina matrix precursor using a sol-gel process. Nextel™ 720 /alumina CMC was made by a vacuum bag process under low pressure and low temperature followed by a pressure-less sintering technique. During processing, numerous matrix cracks, with a porosity of about 26.1%, formed due to the mismatch of thermal expansion coefficient between the matrix and the fibers.

4.2. Boundary conditions

In the actual rapid pneumatic heating process, in the initial state, it is considered that the temperature distribution inside the material is a constant $T(x) = 300$ K. Therefore, the temperature distribution inside the material is also linear, representing the temperature on the lower surface of the material and the temperature on the upper surface of the material; Simplify the heating process as a one-dimensional heat conduction process, where the windward side of the material undergoes forced convection heat transfer with high-temperature gas, and the leeward side is insulated. The heat transfer coefficient and gas temperature on the windward side of the material are known: the heat transfer coefficient is denoted as H , which is related to the surface temperature of the material, and the gas temperature is denoted as T^{gas} , with $T^{\text{gas}}(t) = 1473$ K. The schematic diagram and boundary conditions of the pneumatic heating of the material are shown in Fig. 3.

It should be noted that the gas temperature and surface heat transfer coefficient given here are only used to characterize the macroscopic thermal boundary during the heating process, and may not necessarily meet the actual heat transfer situation of the material during pneumatic heating. In the analysis in the following Section 5, some of the parameters that have already been determined will be changed to reveal their impact on the energy release rate of the material.

4.3. Thermomechanical behaviors

4.3.1. Transient thermal analysis

Fig. 4(a) shows the temperature changes on the windward and leeward surfaces of the material during the heating process. The calculation results obtained by the proposed model are more conservative than those obtained through fluid-thermal coupling calculation. Specifically, the temperature rise on the leeward side is slower due to the adiabatic condition set for the leeward side. The thermal inertia effect leads to a plateau in the temperature history of the leeward side, resulting in a lower final temperature. While the results from fluid-thermal coupling calculation are more aligned with actual operating conditions, it should be noted that determining the forced convection heat transfer coefficient on the leeward side still has some uncertainties. Both the calculation results in this paper and the heat flow coupling calculation results exhibit more significant non-uniformity compared to the experimental results. Consequently, the calculated energy release rate will also be higher than the actual situation observed in experiments.

In Fig. 4(b), the transient temperature distribution evolution of typical layers of CMC with specific characteristics is depicted. This is based on Eq. (8) and considers transient rapid heating. Within 0.5 s after the

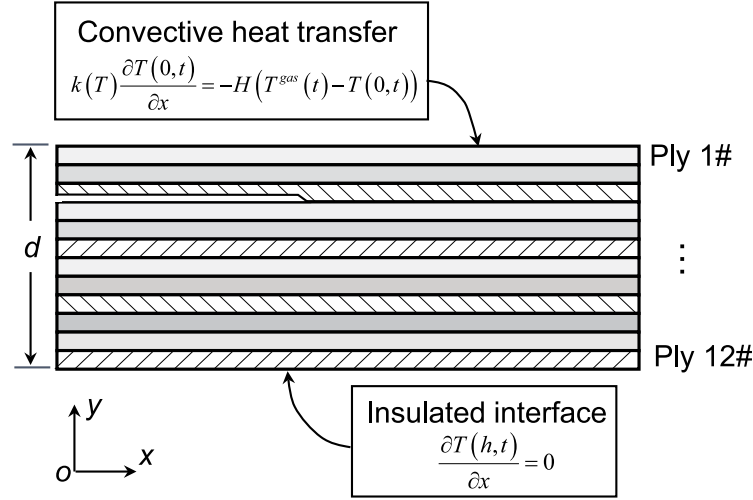


Fig. 3. Schematic of the problem of a multi-layers CMC systems with an insulated interface on the bottom and convective heat transfer on the top.

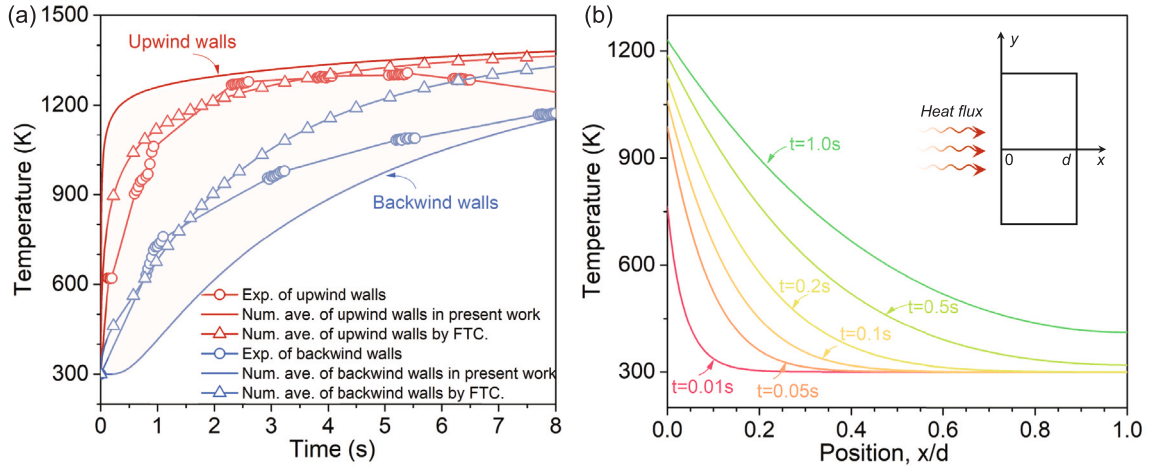


Fig. 4. (a) Temperature history of upwind and backwind walls for the case of $T_{tot} = 1473$ K, and (b) temperature profiles through the multilayers CMC at several times. Note: FTC. represent the fluid-thermal coupling analysis method.

start of transient heating, the temperature on the windward side of the material rapidly rises to 1123 K, while the temperature rise on the leeward side is not significant. During this stage, the high-temperature region in the material exhibits both high temperature and a high-temperature gradient. The overall temperature distribution during this stage is approximately bilinear, but as the heating process progresses, the temperature distribution in the material gradually approaches a linear steady state. The rapid heating stage occurs within 0.5s after the start of heating. At this stage, the significant temperature difference between the high-temperature region and the low-temperature zone can result in substantial thermal stress and energy release rate in the composite, leading to damage such as matrix cracks, interface debonding, and delamination.

4.3.2. Stresses analysis

The transient stress changes under the scenario of CMC structure with transient rapid heating are shown in Fig. 5. Under bending constraints as illustrated in Fig. 5(a), the stress levels on the surface and sub-surface areas of CMC laminates are relatively high. It should be pointed out that the peak tensile stress on the surface of the material occurs at a temperature of 1 s. At this time, the temperature of the neutral layer and below of the material begins to undergo significant changes, but the material remains in a tensile stress state, and the maximum stress value is not high. The cracks generated by this driving

force tend to exhibit Type I (open type) cracks. The variation pattern of stress in each layer of non-destructive material far ahead of the crack tip over time is shown in Fig. 5(b). Due to the rapid heating of the surface relative to other areas of the material, a very large compressive stress is generated in the surface area of the material. The peak of the compressive stress occurs within 0.5s of the heating process. As the temperature inside the material gradually becomes uniform, the temperature gradient gradually decreases, the compressive stress on the surface and sub-surface of the material decreases rapidly. It is interesting that within the first 0.6s of the rapid heating process, due to the low thermal conductivity of the material, the temperature of the sub-surface (located approximately at $x/d = 0.25 \sim 0.33$ of the material) is lower. Therefore, under the constraint of compressive stress on the material surface, this position exhibits a tensile stress state. However, as the temperature of the sub-surface layer of the material increases, their stress state quickly transitions from a tensile state to a compressive state. The rapid transition of this stress state results in the initiation and propagation of cracks in the material not on the surface but at the subsurface.

To reveal the effect of bending constraints on materials, the CMC laminated board is no longer subjected to bending constraints, and its transient stress and position curve over time are shown in Fig. 5(e). In the absence of bending constraints, the maximum surface stress of the material occurs within 0.4 s after the start of cooling. The

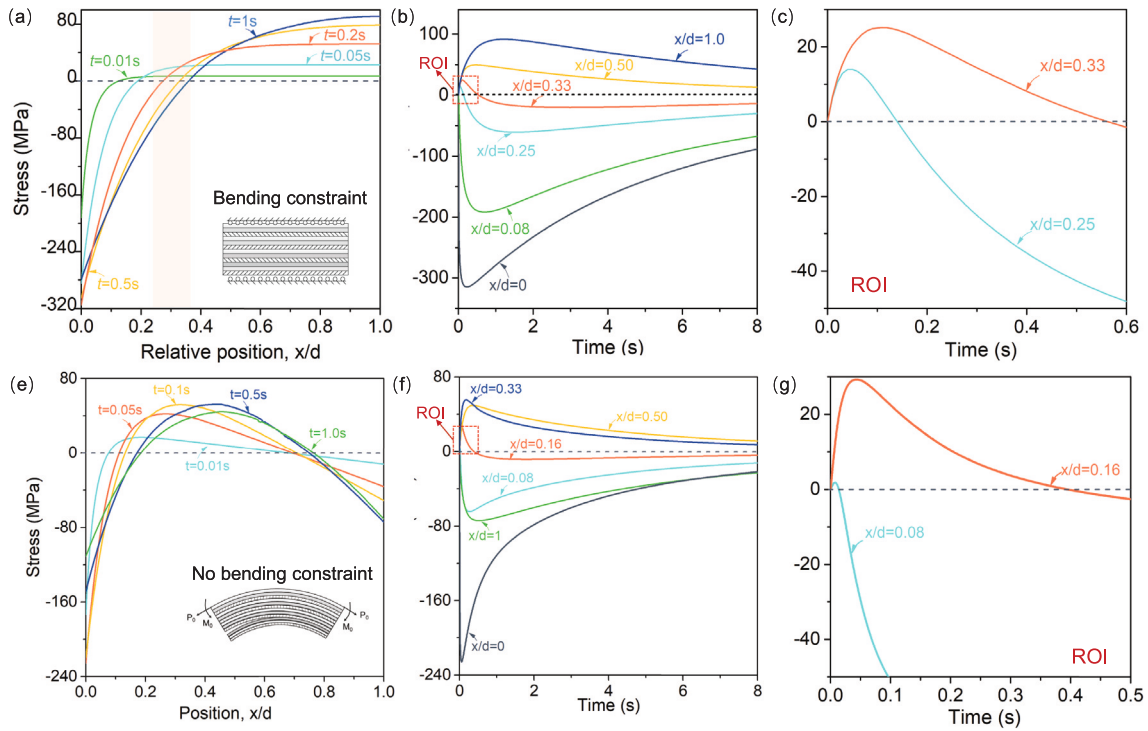


Fig. 5. (a)–(c) Transient stress variations in the CMC laminate for bending constraint. (e)–(g) Transient stress variations in the CMC laminate for no bending constraint.

maximum stress value is only 236 MPa, and then the stress gradually decreases. However, compared to the bending constraint situation, its value is nearly one-third smaller, indicating that the bending constraint effect has a significant impact on the stress distribution of each layer inside the material. Similarly, the stress state on the sub-surface of the material (located approximately at $x/d = 0.08 \sim 0.20$) undergoes a transition from a tensile state to a compressive state, and the time of the transition from tensile to compressive is about 0.05s earlier than that of the bending constraint state, because the tensile and compressive state of the material appears closer to the surface without bending constraint.

In addition, as shown in Fig. 5(f), the bending of the neutral layer and below in the material significantly reduces the stress on the surface and subsurface layers of the material. In contrast, without bending constraints, the stress on the surface and subsurface layers of the material is higher. This seems to suggest that the difference in stress distribution with and without bending strain can lead to significant differences in material energy release rates. The stress distribution depends on the degree to which the CMC laminate undergoes overall bending, which is a function of the material thickness and its constraint method. This indicates that bending constraints play an important regulatory role in the stress distribution of each layer inside the material.

The results indicate that studying the stress distribution and transition state within CMC during rapid heating is crucial for predicting and controlling potential damage initiation and propagation. These findings contribute to the design and engineering of CMC structures, ultimately improving their performance and durability in thermal protection applications.

4.3.3. Energy release rate

The energy release rates of layered cracks under no bending constraint and bending constraint conditions are shown in Fig. 6 and Fig. 7, respectively. It can be seen that due to the rapid heating of the CMC surface and the significant compressive stress related to the surface, the maximum energy release rate of the double-layer bending constraint strain is approximately twice that of no bending constraint case. This important impact is mainly due to the higher stress resistance in CMC laminates to bending.

For bending constraint cases, the maximum energy release rate occurs approximately 1.0 s after the start of cooling (Fig. 6(a)), with the location of delamination cracks ranging from 0.25 to 0.33. For no bending constraint cases, the maximum energy release rate occurs approximately 0.5 s after the start of cooling (Fig. 7(a)), with the location of delamination cracks ranging from 0.08 to 0.2. The advance of the maximum energy release rate and the closer proximity of the delamination crack to the heated surface indicate that bending constraints to some extent limit the overall deformation of the material.

Under the bending constraint, the energy release rate of the material at various positions during the rapid heating process changes as shown in Fig. 6(b), with the maximum energy release rate occurring at the positions of the third and fourth layers. In the absence of bending constraints, the maximum energy release rate occurs at the positions of the first and second layers, as shown in Fig. 6(b), which is closer to the rapidly heated surface. The energy release rate is influenced by both the stress in the material and the thickness of the material on both sides of the delamination position. By comparing the results of Fig. 6 with the calculated stress distribution (Fig. 5) mentioned earlier, it can be observed that the position with the highest energy release rate in the material is where a significant transition between tensile and compressive stress states occurs. In fact, after changing the conditions later and conducting the same analysis, the same results were also obtained.

However, regardless of whether there is bending constraint or no bending constraint, the main type of delamination crack is type II (slip type) crack as demonstrated in Fig. 8. Because the rapidly heated surface exhibits compressive stress, while the neutral layer and below bear a certain amount of tensile stress, their value is smaller than the matrix cracking stress or interface debonding stress of the material. When the entire material gradually reaches thermal equilibrium (about 8s), the energy release rate monotonically decreases to a limit, which is much lower than the maximum value of the asymptotic mode mixing that still has the main component of mode I.

In most cases, the main contribution to the energy release rate of delamination cracks is the elastic energy in the rapidly heated surface

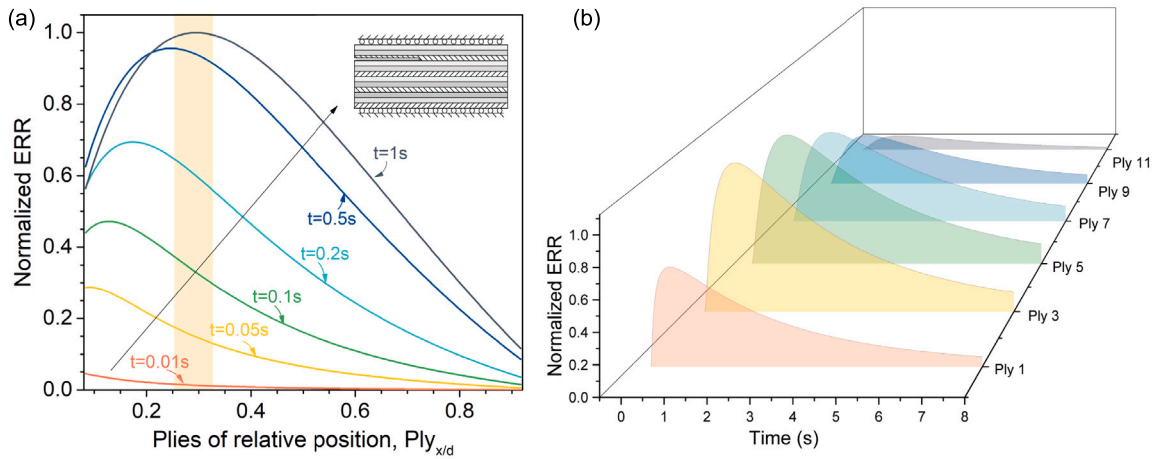


Fig. 6. Transient variation of the delamination energy release rate for bending constraint. The temperature and stress distributions are considered. Additionally, the variation of the energy per unit area in the coating U_{HD} , well ahead of the delamination crack tip, is included for both cases. It is evident that U_{HD} provides a good approximation to G , the delamination energy release rate.

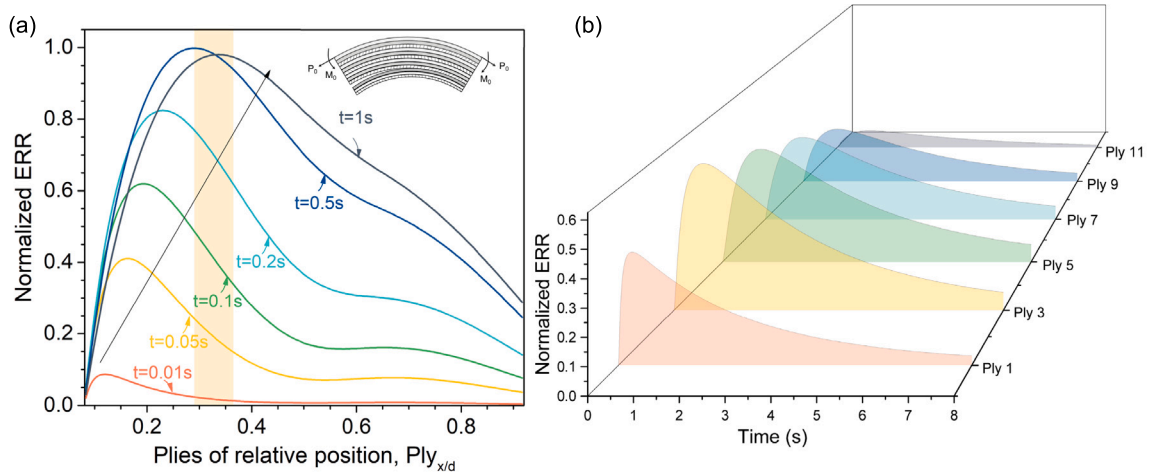


Fig. 7. Transient variation of the delamination energy release rate for no bending constraint.

and subsurface layers of CMC laminates. G is not greater than 10% of the surface and subsurface strain energy. Even without bending constraints, the elastic energy on the surface and subsurface of the material provides the most significant contribution to the rate of energy release. The energy provided by the neutral layer and below is much less. However, not all elastic energy in the surface and subsurface is released when delamination cracks occur, unless the temperature distribution happens to be strictly linear at that moment.

5. Energy release rate as a function of system properties

The rapid aerodynamic heating scenario determined in Section 4.3 is a typical case of damage and failure of ceramic matrix composite laminates under transient thermal loads. It can be seen that the energy release rate depends on a specific material system. Therefore, this section will calculate the energy release rate under different material parameters, including material modulus, material thickness, thermal conductivity, thermal expansion coefficient, and different thermal boundary conditions (surface heat transfer coefficient). To explore their impact on the energy release rate of materials. By conducting comprehensive parameter studies, we aim to gain a deeper understanding of its impact on the performance of ceramic matrix composite laminates during rapid heating processes. This will provide an important reference for designing and manufacturing ceramic matrix composite structures with high performance and reliability.

5.1. The role of the substrate thickness

The effects of material thickness on the energy release rate are illustrated in Fig. 9. Herein, the normalized energy release rate is introduced, defined as the ratio of the energy release rate for a given thickness to the maximum energy release rate observed across all the examined conditions. It can be observed that while keeping the thermal boundary conditions and initial temperature of the material constant, reducing the material thickness can significantly decrease the energy release rate. This can be attributed to two factors. Firstly, as the thickness of the material decreases, the overall stiffness of the material will decrease, resulting in a reduction in stored strain energy. Secondly, as the thickness of the material decreases, the thermal non-uniformity in the material during the heating process will also decrease.

Moreover, it should be noted that a lower material thickness not only results in a lower energy release rate, but also a faster decrease in energy release rate after reaching its peak. This is more significant compared to the decrease in peak value. This result is very meaningful for engineering applications, implying that the thickness of the material structure can be minimized as much as possible while meeting the structural stiffness requirements, which can reduce the initiation of delamination cracks in the material.

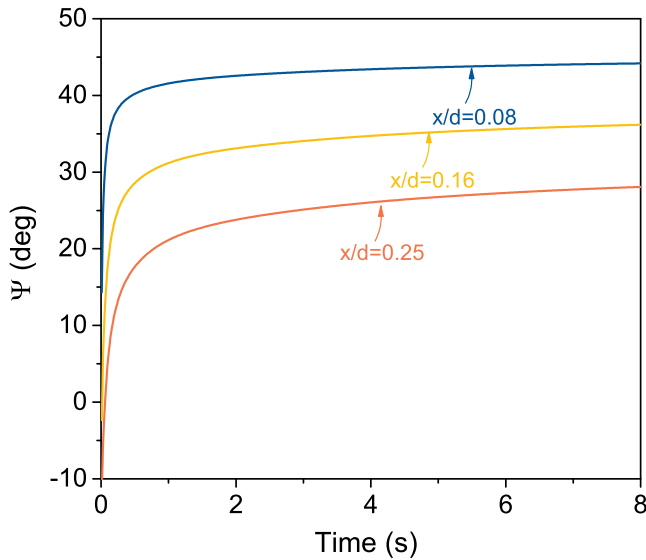


Fig. 8. The change of the crack mode mixity phase angle ψ with time calculate by Eqs. (28)–(29).

5.2. The role of the materials modulus

In the analysis above, it was found that for CMC laminates with a thickness of 3 mm, the maximum energy release rate occurs at the $x/d = 0.25$ position. Based on this observation, the material structure can be optimized by designing the modulus of the material along the thickness direction. This can be achieved by appropriately adjusting the modulus from the windward surface to the sub-surface at $x/d = 0.25$ through the manufacturing process.

We can introduce the concept of gradient modulus, defined as the ratio of the modulus from the windward surface to the sub-surface at $x/d = 0.25$ to the overall elastic modulus ($E_{\text{grad}} = \tilde{E}/E_0$). By examining the effect of changes in the gradient modulus on the energy release rate, we can gain insights into the material's behavior.

Due to the nonlinear constitutive relationship of the material, Fig. 10(a) illustrates the influence of the modulus on the maximum energy release rate in the material while keeping other parameters unchanged. It can be observed that the maximum energy release rate exhibits an approximate linear relationship with the modulus. This indicates that adjusting the modulus through the gradient modulus concept can effectively control and optimize the energy release rate in the material.

5.3. The role of thermal conductivity coefficient

By changing only the thermal conductivity coefficient of the material while keeping all other parameters the same as the original conditions, the maximum energy release rate curve is obtained as shown in Fig. 10(b). Since the thermal conductivity coefficient is a linear function of temperature, controlling the ratio is considered as a method to change the thermal conductivity coefficient. It can be observed that increasing the thermal conductivity coefficient and reducing the material thickness have similar effects on the energy release rate and stress. A higher thermal conductivity coefficient leads to a more uniform temperature distribution in the material during the heating process. As a result, the energy release rate of the material exhibits a lower peak and decreases at a faster rate after reaching the peak.

This suggests that both increasing the thermal conductivity coefficient and reducing the material thickness can effectively control and optimize the energy release rate in the material. These findings provide valuable insights for designing and engineering materials with desired energy release rate characteristics.

5.4. The role of coefficient of thermal expansion

In Fig. 10(c), the effect of the thermal expansion coefficient on the maximum energy release rate in materials is shown. The thermal expansion coefficient of materials is temperature-dependent, and controlling the ratio is used to change the thermal expansion coefficient while keeping other conditions unchanged. Fig. 10(c) illustrated that stresses associated with thermal transients scale with the CTE of the CMC itself, this observation and results that transient-driven cracking scales with coating CTE informs the development of CMC systems, as it implies that attempts to mitigate delamination failures by matching CTE of fiber and matrix address only part of the problem. It can be observed that, similar to the effect of changing the modulus, there is a linear relationship between the ratio of maximum stress and the ratio of material thermal expansion coefficient. However, the ratio of maximum energy release rate maintains a square relationship with the ratio of material thermal expansion coefficient.

From this perspective, changing the thermal expansion coefficient of a material has a more significant effect on reducing the energy release rate compared to changing its modulus. This suggests that adjusting the thermal expansion coefficient through the method of controlling the ratio can be an effective strategy to optimize and reduce the energy release rate in materials. These findings provide valuable insights for material design and engineering applications.

5.5. The role of heat transfer coefficient

Fig. 10(d) illustrated that stresses associated with thermal transients scale with the heat transfer coefficient. The heat transfer coefficient of the windward side has a significant impact on the peak energy release rate during the initial heating stage. Because the higher the heat transfer coefficient on the windward side, the faster the surface heating rate, the greater the temperature rise relative to the rest of the material, the greater the stress on the surface, and the higher the energy release rate. However, as the temperature of the windward surface of the material rapidly increases, the heat transfer on the surface will rapidly decrease. Therefore, after the stress and energy release rate reach their peak, the influence of the surface heat transfer coefficient on these two indicators will gradually weaken until it disappears.

By studying the impact of different material properties, such as thermal conductivity, coefficient of thermal expansion, specific heat capacity, and mechanical properties, it becomes possible to tailor the composition of CMCs to meet specific performance requirements under rapid heating scenarios. Additionally, exploring the influence of layer thickness on thermal response can offer valuable guidance for optimizing the design of composite structures to withstand thermal stress and minimize delamination risks. Furthermore, analyzing the effects of heat transfer conditions can help in developing strategies to enhance the thermal performance and durability of CMC components in practical applications.

6. Conclusions

In the present study, a multi-layer thermomechanical induced crack initiation model for ceramic matrix composite laminates subjected to thermal gradients and transients is proposed, and the main conclusions are as follows.

- During the rapid pneumatic heating process, the non-uniform temperature distribution in the material leads to the generation of stress. A higher temperature gradient corresponds to a greater stress in the material. In certain positions within the material, there may be a transition from a tensile stress state to a compressive stress state.

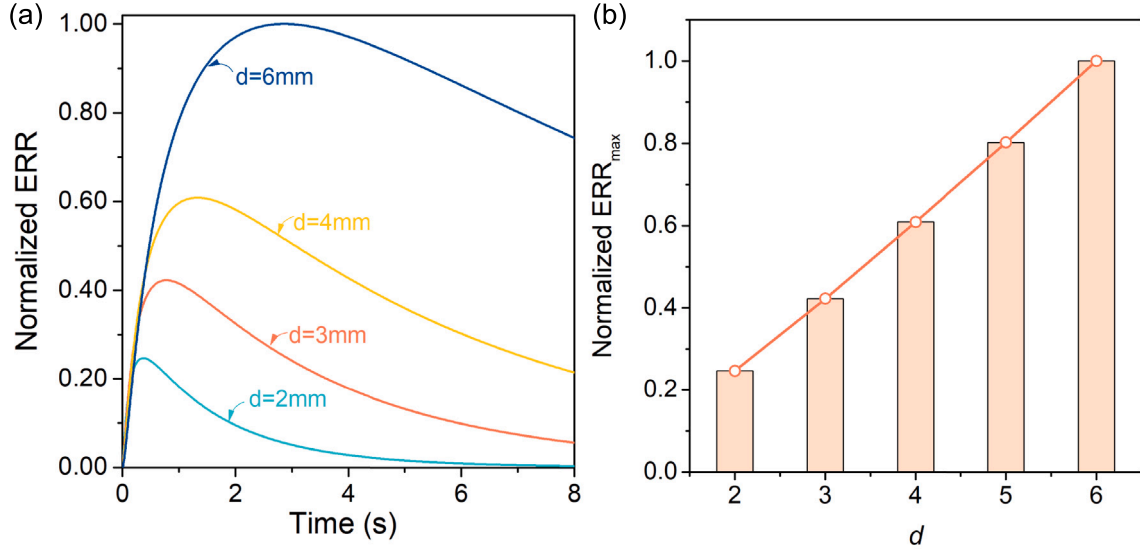


Fig. 9. The effect of thickness on the delamination crack energy release rate for CMC laminate specified by Eq. (27) and subject to the thermal boundary condition in Fig. 3.

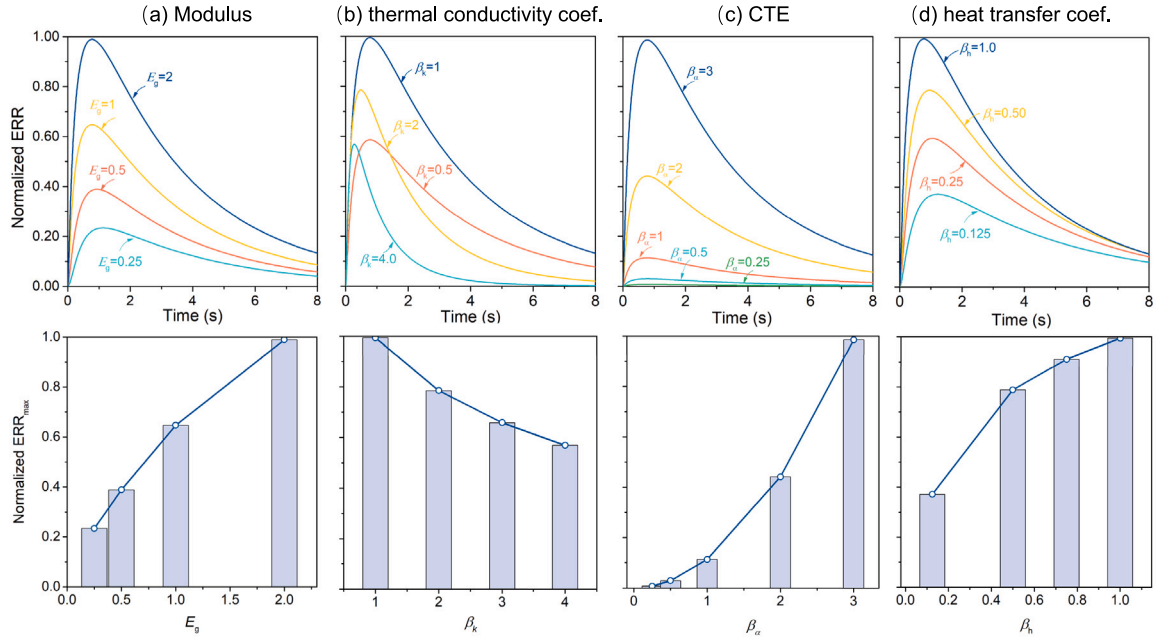


Fig. 10. The effect of the Materials modulus, thermal conductivity coefficient, CTE, and heat transfer coefficient on the delamination crack energy release rate for CMC laminate specified by Eq. (27) and subject to the thermal boundary condition in Fig. 3.

- By improving the material properties near the windward surface, such as enhancing thermal conductivity or reducing thermal expansion coefficient, the stress and energy release rate during the start-up heating process can be effectively reduced. This improvement in material properties can lead to enhanced performance and durability of the material.
- The delamination energy release rate of materials is influenced by both stress and the thickness of the material on both sides of the delamination position. However, based on the calculation results, it is observed that the position with the highest energy release rate highly overlaps with the position where the transition from tensile to compressive stress occurs. This suggests that managing stress and understanding stress distribution is crucial in predicting and controlling the delamination energy release rate in materials.

CRediT authorship contribution statement

Zhengmao Yang: Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Jiangtao Wang:** Formal analysis, Data curation. **Rui Yang:** Resources, Conceptualization. **Jian Jiao:** Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The present work is supported by the National Natural Science Foundation of China (NSFC) (Grant Nos. 52105165).

References

- Alabdullah, M., Ghoniem, N.M., 2020. A thermodynamics-based damage model for the non-linear mechanical behavior of SiC/SiC ceramic matrix composites in irradiation and thermal environments. *Int. J. Damage Mechanics* 29 (10), 1569–1599. <http://dx.doi.org/10.1016/j.commat.2008.09.004>, URL <http://www.sciencedirect.com/science/article/pii/S0927025608004114>.
- Bumgardner, C.H., Heim, F.M., Roache, D.C., Jarama, A., Price, M.C., Deck, C.P., Li, X., 2021. Analysis of SiC/SiC composites for energy applications at ambient conditions. *J. Am. Ceram. Soc.* 104 (1), 481–491. <http://dx.doi.org/10.1111/jace.17462>.
- Di Caprio, F., Russo, A., Manservigi, C., Scigliano, R., De Stefano Fumo, M., Tescione, D., Riccio, A., 2021. Damage tolerance evaluation of a C/C-SiC composite body flap of a re-entry vehicle. *Compos. Struct.* 274, 114341. <http://dx.doi.org/10.1016/j.compstruct.2021.114341>, URL <https://www.sciencedirect.com/science/article/pii/S0263822321008035>.
- Genin, G.M., Hutchinson, J.W., 1997. Composite laminates in plane stress: Constitutive modeling and stress redistribution due to matrix cracking. *J. Am. Ceram. Soc.* 80 (5), 1245–1255. <http://dx.doi.org/10.1111/j.1151-2916.1997.tb02971.x>.
- Glass, D.E., 2008. Ceramic matrix composite (CMC) thermal protection systems (TPS) and hot structures for hypersonic vehicles. In: 15th AIAA International Space Planes and Hypersonic Systems and Technologies Conference. American Institute of Aeronautics and Astronautics.
- Hattiangadi, A., Siegmund, T., 2004. A thermomechanical cohesive zone model for bridged delamination cracks. *J. Mech. Phys. Solids* 52 (3), 533–566. [http://dx.doi.org/10.1016/S0022-5096\(03\)00122-4](http://dx.doi.org/10.1016/S0022-5096(03)00122-4), URL <https://www.sciencedirect.com/science/article/pii/S0022509603001224>.
- Hermann, T., McGilvray, M., Naved, I., 2019. Performance of transpiration-cooled heat shields for reentry vehicles. *AIAA J.* 58 (2), 830–841. <http://dx.doi.org/10.2514/1.J058515>.
- Hutchinson, J.W., Evans, A.G., 2002. On the delamination of thermal barrier coatings in a thermal gradient. *Surf. Coat. Technol.* 149 (2), 179–184. [http://dx.doi.org/10.1016/S0257-8972\(01\)01451-7](http://dx.doi.org/10.1016/S0257-8972(01)01451-7).
- Jackson, R.W., Begley, M.R., 2014. Critical cooling rates to avoid transient-driven cracking in thermal barrier coating (TBC) systems. *Int. J. Solids Struct.* 51 (6), 1364–1374. <http://dx.doi.org/10.1016/j.ijsolstr.2013.12.029>.
- Jackson, R.W., Zaleski, E.M., Poerschke, D.L., Hazel, B.T., Begley, M.R., Levi, C.G., 2015. Interaction of molten silicates with thermal barrier coatings under temperature gradients. *Acta Mater.* 89, 396–407. <http://dx.doi.org/10.1016/j.actamat.2015.01.038>, URL <https://www.sciencedirect.com/science/article/pii/S1359645415000518>.
- Koutsakis, G., Begley, M.R., Hutchinson, J.W., Ghandhi, J.B., 2022. Fracture-based transient thermo-mechanical analysis of reciprocating engine thermal barrier coatings. *Eng. Fract. Mech.* 270, 108568.
- Lu, T.J., Fleck, N.A., 1998. The thermal shock resistance of solids. *Acta Mater.* 46 (13), 4755–4768. [http://dx.doi.org/10.1016/S1359-6454\(98\)00127-X](http://dx.doi.org/10.1016/S1359-6454(98)00127-X).
- Pature, N.P., 2016. Advanced structural ceramics in aerospace propulsion. *Nat. Mater.* 15 (8), 804–809. <http://dx.doi.org/10.1038/nmat4687>.
- Parthasarathy, T.A., Keller, K.A., Mah, T.-I., Kerans, R.J., Butner, S., 2004. Reduction of thermal-gradient-induced stresses in composites using mixed fibers. *J. Am. Ceram. Soc.* 87 (4), 617–625. <http://dx.doi.org/10.1111/j.1551-2916.2004.00617.x>.
- Pichon, T., Barreteau, R., Soyris, P., Foucault, A., Parenteau, J.M., Prel, Y., Guedron, S., 2009. CMC thermal protection system for future reusable launch vehicles: Generic shingle technological maturation and tests. *Acta Astronaut.* 65 (1), 165–176. <http://dx.doi.org/10.1016/j.actaastro.2009.01.035>, URL <https://www.sciencedirect.com/science/article/pii/S0094576509000502>.
- Rajan, V.P., Shaw, J.H., Rossol, M.N., Zok, F.W., 2014. An elastic-plastic constitutive model for ceramic composite laminates. *Composites* 66, 44–57. <http://dx.doi.org/10.1016/j.compositesa.2014.06.013>.
- Shi, T., Zhang, Y., Zhang, X., Wang, Y., Zheng, K., 2023. A strength based thermo-mechanical coupled cohesive zone model for simulating heat flux induced interface debonding. *Compos. Sci. Technol.* 243, 110255. <http://dx.doi.org/10.1016/j.compotech.2023.110255>, URL <https://www.sciencedirect.com/science/article/pii/S0266353823003494>.
- Sundaram, S., Lipkin, D.M., Johnson, C.A., Hutchinson, J.W., 2012. The influence of transient thermal gradients and substrate constraint on delamination of thermal barrier coatings. *J. Appl. Mech.* 80 (1), <http://dx.doi.org/10.1115/1.4007727>.
- Wang, J., Yang, Z., 2024a. Damage evolution in porous matrix oxide fiber-reinforced ceramic composites at high thermal gradient using wind tunnel tests. *AIAA J.* 62 (6), 2331–2341. <http://dx.doi.org/10.2514/1.J063058>.
- Wang, J., Yang, Z., 2024b. High-temperature transient-induced thermomechanical damage of fiber-reinforced ceramic-matrix composites under supersonic wind tunnel tests. *Composites A* 184, 108243. <http://dx.doi.org/10.1016/j.compositesa.2024.108243>.
- Yang, Z., Yuan, H., 2019. Evolution and characterization of cyclic thermal shock-induced thermomechanical damage in oxide/oxide ceramics matrix composites. *Int. J. Fatigue* 120, 150–161. <http://dx.doi.org/10.1016/j.ijfatigue.2018.11.006>, URL <https://www.sciencedirect.com/science/article/pii/S0142112318306017>.
- Zok, F.W., Collier, V.E., Begley, M.R., 2021. Coating recession effects in ceramic composite strength. *J. Mech. Phys. Solids* 156, 104608. <http://dx.doi.org/10.1016/j.jmps.2021.104608>, URL <https://www.sciencedirect.com/science/article/pii/S0022509621002519>.